

THE STRICT DIRICHLET PÓLYA INEQUALITY ON SIMPLY CONNECTED PLANAR DOMAINS

ABSTRACT. We prove that the Dirichlet eigenvalues of every bounded simply connected planar domain satisfy $|\Omega|\lambda_j(\Omega) > 4\pi j$ for all $j \geq 1$, without any boundary regularity assumption. The proof associates with conformal boundary curves a unitary path with positive trace-class generator, injective at every positive radius, and integrated trace equal to one half of the energy times the area. A boundary identity transfers Dirichlet resonances to unitary eigenphases approaching a full turn. A Ky Fan estimate controls any prescribed number of crossings by the corresponding partial eigenvalue sum of the generator. The remaining spectral tail is strictly positive and survives conformal exhaustion.

1. INTRODUCTION

Let $G \subset \mathbb{R}^2$ be a bounded nonempty open set. The Dirichlet Laplacian $-\Delta_G^D$ is the self-adjoint operator in $L^2(G)$ associated with the quadratic form

$$q_G[u] = \int_G |\nabla u|^2, \quad \text{Dom } q_G = H_0^1(G).$$

We denote its eigenvalues, repeated according to multiplicity, by $\lambda_1(G) \leq \lambda_2(G) \leq \dots$, and write

$$N_G(E) = \#\{j : \lambda_j(G) \leq E\}.$$

We set $N_G(E) = 0$ for $E \leq 0$. All counting functions in this paper include the eigenvalues at the stated energy.

For a bounded planar domain with smooth boundary, Weyl's law [13] gives

$$N_G(E) = \frac{|G|}{4\pi} E + o(E) \quad (E \rightarrow \infty).$$

Pólya asked whether the leading term is an upper bound at every energy, rather than only an asymptotic approximation. The conjecture appears in his 1954 work on plausible reasoning [11]. In its planar Dirichlet form, it asserts that

$$\lambda_j(G) \geq \frac{4\pi j}{|G|}, \quad j \geq 1.$$

The strict form replaces this inequality by $>$ at every index.

Pólya proved the non-strict inequality for domains that tile the plane [12]. The sharp inequalities of Berezin [1] and Li–Yau [9] control spectral averages on general domains, but do not give the conjectured bound for each eigenvalue. Freitas and Salavessa [7] obtained families of non-tiling examples. Filonov, Levitin, Polterovich, and Sher proved the conjecture for disks and arbitrary circular sectors, and the Dirichlet conjecture for Euclidean balls [5]. They subsequently treated planar annuli [6]. These results exploit special geometry; they do not supply the inequality for an arbitrary simply connected planar domain.

Our main result gives the strict inequality for all bounded simply connected planar domains, without a regularity assumption on the boundary.

Theorem 1.1. *Let $\Omega \subset \mathbb{R}^2$ be a bounded nonempty simply connected domain. Its Dirichlet eigenvalues satisfy*

$$(1.1) \quad |\Omega| \lambda_j(\Omega) > 4\pi j \quad (j \geq 1).$$

Its counting function satisfies

$$(1.2) \quad N_\Omega(E) < \frac{|\Omega|E}{4\pi} \quad (E > 0).$$

Here is the mechanism of the proof. At a fixed energy k^2 , we transport the unilateral shift along conformal boundary curves. After correcting for the moving initial point and reversing the resulting unitary operator, we obtain a path U_r whose generator $L_r = -iU_r^* \partial_r U_r$ is positive and trace class. Its integrated trace is exactly

$$\int_0^1 \text{tr } L_r dr = \frac{k^2}{2} |\Omega|.$$

An identity involving the Dirichlet-to-Neumann map relates a Dirichlet eigenvalue to eigenvalues of U_r approaching 1 through the lower half of the unit circle. Counting crossings of $e^{i\alpha}$ for $\alpha < 2\pi$ and then letting $\alpha \uparrow 2\pi$ recovers the coefficient 4π .

Strictness requires more than this full-trace estimate. If at least m crossings occur, we bound their contribution using only the sum of the m largest eigenvalues of L_r . The difference between the full trace and this partial sum is positive, since L_r is injective on an infinite-dimensional space for every $r > 0$. We retain that difference through the limit in α and through conformal exhaustion. This gives the strict inequality even when the boundary is not regular.

The distinction between an eigenvalue approaching 1 and an actual eigenvalue at 1 is essential. A related distinction occurs in the inside–outside spectral duality of Eckmann and Pillet [3]. We do not identify the boundary transport used here with their scattering matrix. Instead, we prove the required boundary identity directly on traces of Herglotz waves. This also specifies the domain of the unbounded Cayley transform. The density argument is classical in scattering theory; see Colton and Kress [2]. We retain its short proof in the precise trace topology needed below.

Section 2 gives the eigenphase estimate with a prescribed number of crossings. Section 3 constructs the boundary transport and proves the trace identity and positivity of the spectral tail. Section 4 establishes the boundary identity on Herglotz traces. Section 5 relates Dirichlet eigenvalues to eigenphases near 2π . Section 6 completes the counting argument and passes the positive remainder to arbitrary simply connected domains.

Throughout, inner products are conjugate-linear in the first variable. We identify $\mathbb{R}^2$ with $\mathbb{C}$ and put $\mathbb{D} = \{z : |z| < 1\}$ and $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. Boundary spaces on $\mathbb{T}$ use $d\theta$; the direction space of Herglotz waves uses $d\phi/(2\pi)$. The pairing $\langle h, \Lambda h \rangle$ is the $H^{1/2}$ – $H^{-1/2}$ duality pairing, consistent with the $L^2(d\theta)$ inner product for smooth data. If h_n are the coefficients in the orthonormal basis $(2\pi)^{-1/2} e^{in\theta}$, our Sobolev norm is $\|h\|_{H^s(\mathbb{T})}^2 = \sum_{n \in \mathbb{Z}} (1 + n^2)^s |h_n|^2$. We write $\mathcal{B}(\mathcal{H})$ for the bounded operators on $\mathcal{H}$, and I for the identity. The symbols σ , $\ker$, ran , rank , and Dom denote spectrum, kernel, range, rank, and domain. The symbols $\mathcal{S}_1$ and $\mathcal{S}_2$ denote the trace-class and Hilbert–Schmidt ideals, respectively, and tr denotes the trace. For bounded operators, $[A, B] = AB - BA$.

2. EIGENPHASE ESTIMATES FOR POSITIVE UNITARY PATHS

For a positive trace-class operator A , let $\mu_1(A) \geq \mu_2(A) \geq \cdots \geq 0$ be its eigenvalues, repeated according to multiplicity and supplemented by zeros when necessary. For every integer $m \geq 1$, set

$$(2.1) \quad \Phi_m(A) = \sum_{\ell=1}^m \mu_\ell(A), \quad \tau_m(A) = \text{tr } A - \Phi_m(A).$$

We use Ky Fan’s variational principle [4] in the form

$$(2.2) \quad \Phi_m(A) = \sup\{\text{tr}(PA) : P = P^* = P^2, \text{rank } P \leq m\}.$$

In particular, for positive trace-class A and B ,

$$(2.3) \quad |\Phi_m(A) - \Phi_m(B)| \leq \|A - B\|_{\mathcal{S}_1}, \quad |\tau_m(A) - \tau_m(B)| \leq 2\|A - B\|_{\mathcal{S}_1}.$$

Indeed, the first bound follows by taking the supremum in (2.2) and using $|\operatorname{tr}(P(A-B))| \leq \|A-B\|_{\mathcal{S}_1}$; the second also uses continuity of the trace. If A is injective on an infinite-dimensional space, then $\tau_m(A) > 0$ for every finite m .

For $0 < \alpha < 2\pi$, a crossing of $e^{i\alpha}$ is a time at which it is an eigenvalue of the unitary path. Its multiplicity is the dimension of that eigenspace. All such crossings in the next lemma are isolated and have increasing local eigenphases.

Lemma 2.1. *Let $U : [0, b] \rightarrow \mathcal{B}(\mathcal{H})$ be a continuously differentiable unitary path on a separable Hilbert space, with $U(0) = I$. Suppose*

$$L(t) = -iU(t)^*U'(t) \geq 0$$

is trace class and trace-norm continuous. Assume that U is locally real analytic in operator norm for $t > 0$, and that $\langle v, L(t)v \rangle > 0$ for every $v \neq 0$ and $t > 0$. Set $B = \int_0^b \operatorname{tr} L(t) dt$. Then $U(t) - I$ is trace class, and the following statements hold.

(i) *For $0 < \alpha < 2\pi$, the number c_α of crossings of $e^{i\alpha}$ in $(0, b)$, counted with multiplicity, is finite, and all crossings are positive. For every integer m with $1 \leq m \leq c_\alpha$,*

$$(2.4) \quad \alpha m \leq \int_0^b \Phi_m(L(t)) dt.$$

In particular,

$$(2.5) \quad \alpha c_\alpha \leq B.$$

(ii) *At each fixed time, $U(t)$ has only finitely many eigenvalues in $\{e^{i\theta} : \pi < \theta < 2\pi\}$, with multiplicity.*

(iii) *Let $0 < a < b_1 \leq b$ and $\pi < \alpha < 2\pi$. Suppose $U(a)$ has no spectrum with argument in $[\alpha, 2\pi)$, $e^{i\alpha} \notin \sigma(U(b_1))$, and $U(b_1)$ has at least m eigenvalues with arguments in $(\alpha, 2\pi)$. There are at least m crossings of $e^{i\alpha}$ in (a, b_1) .*

Proof. Since $U' = iUL$, integration in trace norm gives $U(t) - I \in \mathcal{S}_1$. Every spectral point other than 1 is therefore an isolated eigenvalue of finite multiplicity. Near such a point, analytic Riesz projections and a local logarithm reduce the spectral problem to an analytic Hermitian matrix family; we use the standard perturbation results in [8, Chapters II and VII]. For a normalized eigenvector branch with eigenvalue $e^{i\theta(t)}$,

$$\theta'(t) = \langle v(t), L(t)v(t) \rangle.$$

At a multiple eigenvalue, the eigenphase derivatives are the eigenvalues of the compression of $L(t)$ to the eigenspace. This compression is positive definite. Hence every crossing away from 1 is locally isolated and positive.

Choose finite-rank orthogonal projections $P_n \uparrow I$ strongly, put $L_n = P_n L P_n$, and let U_n be the unitary solution on $P_n \mathcal{H}$ of $U'_n = iU_n L_n$, $U_n(0) = I$; extend U_n by the identity on $P_n^\perp \mathcal{H}$. The image of $[0, b]$ under L is compact in $\mathcal{S}_1$, so

$$\sup_t \|L_n(t) - L(t)\|_{\mathcal{S}_1} \longrightarrow 0.$$

Moreover,

$$\frac{d}{dt}(U_n U^*) = iU_n(L_n - L)U^*.$$

Integration in trace norm therefore gives

$$(2.6) \quad \sup_t \|U_n(t) - U(t)\|_{\mathcal{S}_1} \leq \int_0^b \|L_n(t) - L(t)\|_{\mathcal{S}_1} dt \longrightarrow 0.$$

The differential equations also give uniform operator-norm convergence of the first derivatives.

In finite dimension, the eigenphases admit continuous, absolutely continuous, nondecreasing lifts $\theta_{n,j}$ starting at zero. They may be obtained from local Hermitian logarithms and continued along the interval. At almost every time their derivatives are diagonal entries of $L_n(t)$ in an orthonormal eigenbasis; at a multiple eigenvalue one diagonalizes its compression. In particular,

$$\sum_j \theta_{n,j}(b) = \int_0^b \operatorname{tr} L_n(t) dt \leq B.$$

Fix any m crossings of $e^{i\alpha}$, counted with multiplicity. Let q_j be the number of selected crossings belonging to the j th lifted phase, and put $J = \{j : q_j > 0\}$. Then $\sum_{j \in J} q_j = m$ and $|J| \leq m$. Monotonicity gives

$$\theta_{n,j}(b) \geq \alpha + 2\pi(q_j - 1) \geq \alpha q_j \quad (j \in J).$$

This remains true when the selected crossings omit earlier crossings of the same branch. At almost every time, the eigenvectors belonging to the branches in J form an orthonormal family. The variational principle (2.2) therefore yields

$$(2.7) \quad \alpha m \leq \sum_{j \in J} \theta_{n,j}(b) = \int_0^b \sum_{j \in J} \theta'_{n,j}(t) dt \leq \int_0^b \Phi_m(L_n(t)) dt \leq B.$$

Taking all crossings proves the finite-dimensional bound B/α . More generally, (2.7) applies whenever at least m crossings are available; they need not belong to distinct branches.

We explain the passage to U without following eigenvalues through 1. Fix a crossing time $t_0 \in (0, b)$ of multiplicity q . Choose a small arc about $e^{i\alpha}$, disjoint from 1, whose spectral projection at t_0 has rank q . On a sufficiently small closed time interval about t_0 , that projection keeps rank q . After identifying its ranges with a fixed space, the local logarithm relative to $e^{i\alpha}$ is negative definite at the left endpoint and positive definite at the right endpoint. By (2.6), the corresponding projections for U_n have the same rank, and the endpoint logarithms have the same inertias for large n . Their eigenvalues must therefore cross zero with total signed multiplicity q . The positivity of L_n on $P_n \mathcal{H}$ for positive times makes these crossings positive. The same construction applies to any finite collection of crossing times, using disjoint time intervals. Their total multiplicity is at most B/α . This proves both finiteness and (2.5). If $m \leq c_\alpha$, choose finitely many crossing windows containing at least m crossings of U , with multiplicity. For all sufficiently large n the path U_n has at least m crossings in those windows. Apply (2.7) to m of them and use (2.3) and the uniform trace-norm convergence of L_n to L . This proves (2.4). Both estimates remain valid for interior crossings when an endpoint of the full path is itself a crossing.

At a fixed time, any finite collection of eigenvalues of $U(t)$ in the open lower semicircle persists, with multiplicity, under (2.6). Each corresponding eigenvalue of $U_n(t)$ has a nonnegative lifted phase greater than π . The collection therefore has size at most B/π . Since the collection was arbitrary, this proves (ii).

For (iii), choose $\beta < 2\pi$ above the arguments of m selected eigenvalues at b_1 , with $e^{i\beta}$ outside both endpoint spectra. The spectral projection onto the arc (α, β) has finite rank. Between crossings of the endpoints its rank is constant. Part (i) gives only finitely many endpoint crossings, and their positive directions give the rank balance

$$n_{(\alpha, \beta)}(b_1) - n_{(\alpha, \beta)}(a) = c_\alpha(a, b_1) - c_\beta(a, b_1).$$

Here n denotes spectral multiplicity in the indicated arc, and c denotes crossing multiplicity in the indicated time interval. The initial rank is zero, the final rank is at least m , and $c_\beta(a, b_1) \geq 0$. This proves (iii). $\square$

3. BOUNDARY TRANSPORT AND THE AREA IDENTITY

Until Section 6, assume that $\Omega = F(\mathbb{D})$, where F is holomorphic and injective on a neighborhood of $\overline{\mathbb{D}}$. Fix $k > 0$, and define

$$\Omega_r = F(r\mathbb{D}), \quad \gamma_r(\theta) = F(re^{i\theta}), \quad 0 < r \leq 1.$$

The curves are positively oriented. We write $\gamma'_r = \partial_\theta \gamma_r$; radial derivatives of operators will be written as ∂_r .

Let $\mathcal{H} = \ell^2(\mathbb{N}_0)$, where $\mathbb{N}_0 = \{0, 1, \dots\}$, with standard basis $(e_n)_{n \geq 0}$, and let $Se_n = e_{n+1}$. If $P_0 v = e_0 \langle e_0, v \rangle$, then $[S^*, S] = P_0$. Define the unitary transport $W_r(\theta)$ by

$$(3.1) \quad \partial_\theta W_r = -\frac{ik}{2}(\gamma'_r S^* + \overline{\gamma'_r} S)W_r, \quad W_r(0) = I, \quad V_r = W_r(2\pi).$$

Writing $\gamma_r = x + iy$, set

$$X = -\frac{ik}{2}(S + S^*), \quad Y = \frac{k}{2}(S^* - S).$$

These are bounded skew-adjoint operators, and $[X, Y] = ik^2 P_0/2$. Put

$$C_\theta = x_\theta X + y_\theta Y, \quad C_r = x_r X + y_r Y, \quad B_r = C_r(r, 0) = C_r(r, 2\pi).$$

The moving initial point is removed by solving $\partial_r R_r = B_r R_r$, $R_0 = I$, and setting

$$(3.2) \quad H_r = R_r^* V_r R_r, \quad U_r = H_r^*.$$

The loop at $r = 0$ is constant, so $U_0 = I$.

Proposition 3.1. *The path U_r satisfies the assumptions of Lemma 2.1. Its generator $L_r = -iU_r^* \partial_r U_r$ satisfies, for $0 < b \leq 1$,*

$$(3.3) \quad \int_0^b \text{tr } L_r dr = \frac{k^2}{2} |\Omega_b|.$$

For every integer $m \geq 1$ and every $r > 0$,

$$(3.4) \quad \tau_m(L_r) > 0.$$

Writing $L_{r,k}$ when the energy parameter varies, the map $(r, k) \mapsto L_{r,k}$ is jointly continuous in trace norm for $r \in [0, 1]$ and $k > 0$.

Proof. Differentiating (3.1) yields

$$\partial_\theta(W_r^* \partial_r W_r - W_r^* C_r W_r) = W_r^* (\partial_r C_\theta - \partial_\theta C_r + [C_\theta, C_r]) W_r.$$

The mixed derivatives cancel. Since

$$J = x_r y_\theta - x_\theta y_r = r |F'(re^{i\theta})|^2, \quad [C_\theta, C_r] = -J[X, Y],$$

integration around the boundary gives

$$V_r^* \partial_r V_r = V_r^* B_r V_r - B_r - \frac{ik^2}{2} \int_0^{2\pi} J W_r^* P_0 W_r d\theta.$$

Differentiating (3.2) cancels the first two terms:

$$(3.5) \quad H_r^* \partial_r H_r = -iQ_r, \quad Q_r = \frac{k^2}{2} R_r^* \left(\int_0^{2\pi} J W_r^* P_0 W_r d\theta \right) R_r.$$

Thus $L_r = H_r Q_r H_r^* \geq 0$. The integral in (3.5) is a trace-norm integral of positive rank-one operators. Taking the trace in (3.5) gives

$$\text{tr } L_r = \frac{k^2}{2} \int_0^{2\pi} r |F'(re^{i\theta})|^2 d\theta.$$

Integration from 0 to b proves (3.3) by the conformal area formula. The same representation gives trace-norm continuity, including at $r = 0$.

Suppose $r > 0$ and $\langle v, Q_r v \rangle = 0$. The zeroth component of $f(\theta) = W_r(\theta)R_r v$ vanishes for every θ , since $J > 0$. Its differential equation is $0 = f'_0 = -(ik/2)\gamma'_r f_1$. Since $\gamma'_r \neq 0$, we obtain $f_1 = 0$. Successively applying the component equations gives $f_n = 0$ for every n , and hence $v = 0$. Unitary conjugation proves the required strict positivity of L_r . In particular, L_r is injective on the infinite-dimensional space $\mathcal{H}$. Its compactness then implies (3.4) by (2.1).

The coefficients of both operator differential equations are locally real analytic in r , uniformly in θ . The uniformly convergent Picard series on compact parameter neighborhoods gives operator-norm real analyticity of W_r, V_r, R_r, U_r . At $r = 0$ the same equations and (3.5) give continuous differentiability. This verifies the remaining assumptions of Lemma 2.1.

Continuous dependence in the same differential equations gives joint operator-norm continuity of $W_r(\theta)$, R_r , and H_r in (r, k) , uniformly in θ on compact parameter sets. The integrand in (3.5) is a continuous rank-one operator in trace norm. That formula and $L_r = H_r Q_r H_r^*$ give the stated joint trace-norm continuity, including at $r = 0$ where $J = 0$. $\square$

4. HERGLOTZ WAVES AND THE BOUNDARY IDENTITY

4.1. Density of boundary traces. For $a \in L^2(\mathbb{T}, d\phi/(2\pi))$, define the Herglotz wave

$$(4.1) \quad u_a(z) = \int_0^{2\pi} a(\phi) \exp \left[-\frac{ik}{2}(ze^{-i\phi} + \bar{z}e^{i\phi}) \right] \frac{d\phi}{2\pi}.$$

It is smooth on $\mathbb{R}^2$ and satisfies $(\Delta + k^2)u_a = 0$. We use the following standard consequence of uniqueness for the exterior Helmholtz problem; compare [2, 3].

Lemma 4.1. *Let G be a bounded smooth Jordan domain. If $k^2 \notin \sigma(-\Delta_G^D)$, the traces of the waves (4.1) are dense in $H^{1/2}(\partial G)$.*

Proof. Let a continuous linear functional represented by $g \in H^{-1/2}(\partial G)$ annihilate all the traces in question. Form the outgoing single-layer potential with density g and fundamental solution $(i/4)H_0^{(1)}(k|x - y|)$, where $H_0^{(1)}$ is the Hankel function of the first kind and order zero. The far field is, up to a nonzero constant, the pairing of g with plane waves. The annihilation assumption for arbitrary L^2 direction densities makes this far field zero in L^2 and then in every direction by continuity.

Rellich's uniqueness theorem, or the outgoing Fourier–Hankel expansion outside a large disk, makes the potential zero there. Unique continuation and connectedness of the exterior make it zero throughout $\mathbb{R}^2 \setminus \bar{G}$. The single-layer mapping $H^{-1/2}(\partial G) \rightarrow H_{\text{loc}}^1(\mathbb{R}^2)$ ensures that the interior and exterior Dirichlet traces agree; see [3, Section 3]. Thus the interior solution has zero boundary trace and is zero by the nonresonance assumption. Applying $-\Delta - k^2$ distributionally now gives $g\delta_{\partial G} = 0$, and hence $g = 0$. The Hahn–Banach theorem proves the assertion. $\square$

Smooth boundary parametrizations preserve $H^{1/2}$, so the lemma also applies to the traces pulled back to $\mathbb{T}$. We call a radius r nonresonant when $k^2 \notin \sigma(-\Delta_{\Omega_r}^D)$.

4.2. The Cayley transform.

Lemma 4.2. *At every nonresonant radius, $\ker(V_r - I) = \{0\}$.*

Proof. Suppress r . Embed $\mathcal{H}$ as the nonnegative Fourier modes in $\ell^2(\mathbb{Z}) = L^2(\mathbb{T}, d\phi/(2\pi))$, and let Z be multiplication by $\zeta = e^{i\phi}$. The full-space transport is

$$M(\theta) = \exp \left[-\frac{ik}{2}((\gamma(\theta) - \gamma(0))Z^* + (\overline{\gamma(\theta)} - \overline{\gamma(0)})Z) \right].$$

Its generators commute, and $M(0) = M(2\pi) = I$. Extend $f(\theta) = W(\theta)v$ by zero to the negative indices. Then

$$f' = -\frac{ik}{2}(\gamma'Z^* + \overline{\gamma'}Z)f + \frac{ik}{2}\gamma'f_0e_{-1}.$$

Variation of constants in the bilateral space gives

$$(V - I)v = \frac{ik}{2} \int_0^{2\pi} M(\theta)^* e_{-1} \gamma'(\theta) f_0(\theta) d\theta.$$

If $Vv = v$, multiplication by ζ and by a nonvanishing direction-dependent phase yields

$$\int_0^{2\pi} \gamma'(\theta) f_0(\theta) e^{ik \operatorname{Re}(\gamma(\theta)e^{-i\phi})} d\theta = 0.$$

This equality holds for every direction by continuity. Changing a direction to its opposite shows that the continuous linear functional $h \mapsto \int \gamma' f_0 h d\theta$ annihilates all Herglotz traces. Lemma 4.1 gives $\gamma' f_0 = 0$. The component equations in (3.1) then give $f_n = 0$ successively, so $v = 0$. $\square$

At a nonresonant radius define

$$(4.2) \quad \mathcal{K}_r = i(I + V_r)(I - V_r)^{-1}, \quad \operatorname{Dom} \mathcal{K}_r = \operatorname{ran}(I - V_r).$$

Normality of V_r and Lemma 4.2 make this domain dense. The usual Cayley-transform construction gives a self-adjoint operator. Directly, its two nonreal resolvents are

$$(\mathcal{K}_r + i)^{-1} = \frac{I - V_r}{2i}, \quad (\mathcal{K}_r - i)^{-1} = \frac{(I - V_r)V_r^*}{2i}.$$

They are compact by Lemma 2.1 and Proposition 3.1, so $\mathcal{K}_r$ has compact resolvent. By (3.2), an eigenvalue $e^{i\theta}$ of U_r , with $0 < \theta < 2\pi$, corresponds, with the same multiplicity, to

$$(4.3) \quad \cot(\theta/2) \in \sigma(\mathcal{K}_r).$$

Lemma 2.1(ii) implies that $\mathcal{K}_r$ has only finitely many negative eigenvalues. At each fixed nonresonant radius there is therefore a gap in the spectrum of U_r immediately below argument 2π . No uniform gap in r is being asserted.

4.3. A Dirichlet-to-Neumann identity. Define $\mathcal{O}_r : \mathcal{H} \rightarrow L^2(\mathbb{T}, d\theta)$ by

$$(\mathcal{O}_r v)(\theta) = \gamma'_r(\theta) \langle e_0, W_r(\theta) v \rangle.$$

Its adjoint and Hilbert–Schmidt norm are

$$(4.4) \quad \mathcal{O}_r^* h = \int_0^{2\pi} W_r(s)^* e_0 \overline{\gamma'_r(s)} h(s) ds, \quad \|\mathcal{O}_r\|_{\mathcal{S}_2}^2 = \int_0^{2\pi} |\gamma'_r|^2 d\theta.$$

Also define the Volterra integral operator

$$(\mathcal{T}_r h)(\theta) = \int_0^\theta \gamma'_r(\theta) \overline{\gamma'_r(s)} \langle e_0, W_r(\theta) W_r(s)^* e_0 \rangle h(s) ds.$$

Comparing the kernels on the two triangles gives

$$(4.5) \quad \mathcal{T}_r + \mathcal{T}_r^* = \mathcal{O}_r \mathcal{O}_r^*.$$

These operators are bounded locally uniformly in $r > 0$. The cut at $\theta = 0$ causes no difficulty for these L^2 operators.

At a nonresonant radius, let $\Lambda_r : H^{1/2}(\mathbb{T}) \rightarrow H^{-1/2}(\mathbb{T})$ be the Dirichlet-to-Neumann map with conormal density:

$$(\Lambda_r h)(\theta) = |\gamma'_r(\theta)| \partial_\nu u(\gamma_r(\theta)), \quad (\Delta + k^2)u = 0, \quad u \circ \gamma_r = h,$$

where ν is the outward unit normal. Conformal pullback identifies Λ_r with the disk Dirichlet-to-Neumann map for

$$(4.6) \quad \mathcal{A}_r = -\Delta - k^2 w_r, \quad w_r(z) = r^2 |F'(rz)|^2.$$

Its Dirichlet realization A_r has domain $H^2(\mathbb{D}) \cap H_0^1(\mathbb{D})$. We distinguish the differential expression $\mathcal{A}_r$ from the operator A_r on that domain.

Put

$$\mathfrak{r}_r[h] = \langle h, i\partial_\theta h \rangle + \frac{ik^2}{4} \langle h, (\mathcal{T}_r - \mathcal{T}_r^*)h \rangle.$$

This is a Hermitian quadratic form on $H^{1/2}(\mathbb{T})$.

Proposition 4.3. *At a nonresonant radius, every Herglotz trace $h = u_a \circ \gamma_r$ satisfies $\mathcal{O}_r^* h \in \text{Dom } \mathcal{K}_r$ and*

$$(4.7) \quad \Lambda_r h = i\partial_\theta h + \frac{ik^2}{4} (\mathcal{T}_r - \mathcal{T}_r^*)h + \frac{k^2}{4} \mathcal{O}_r \mathcal{K}_r \mathcal{O}_r^* h.$$

For every compact interval $I \subset (0, 1]$, there is a constant C_I such that

$$(4.8) \quad |\mathfrak{r}_r[h]| \leq C_I \|h\|_{H^{1/2}(\mathbb{T})}^2 \quad (r \in I, h \in H^{1/2}(\mathbb{T})).$$

Proof. Write $\partial = (\partial_x - i\partial_y)/2$ and $\bar{\partial} = (\partial_x + i\partial_y)/2$, and set

$$f_n(z) = (2i/k)^{n+1} \partial^{n+1} u_a(z), \quad n \geq 0.$$

These are the Fourier coefficients of indices $n+1$ of $a(\phi)e^{-ik \text{Re}(ze^{-i\phi})}$. Parseval's identity gives $f(z) \in \mathcal{H}$ and $\|f(z)\| \leq \|a\|_{L^2(d\phi/(2\pi))}$. Differentiation is valid in $\mathcal{H}$, because the differentiated multipliers are bounded on the direction space. We have

$$\partial f_n = -\frac{ik}{2} f_{n+1}, \quad \bar{\partial} f_n = -\frac{ik}{2} f_{n-1}, \quad f_{-1} = u_a.$$

On the boundary this gives

$$\partial_\theta f = -\frac{ik}{2} (\gamma_r' S^* + \overline{\gamma_r'} S) f - \frac{ik}{2} \overline{\gamma_r'} h e_0.$$

If $v = f(\gamma_r(0))$, variation of constants gives

$$f(\theta) = W_r(\theta)v - \frac{ik}{2} W_r(\theta) \int_0^\theta W_r(s)^* e_0 \overline{\gamma_r'(s)} h(s) ds.$$

Periodicity and (4.4) imply

$$(4.9) \quad (I - V_r)v = -\frac{ik}{2} V_r \mathcal{O}_r^* h.$$

In particular,

$$\mathcal{O}_r^* h = \frac{2i}{k} (V_r^* - I)v = \frac{2i}{k} (I - V_r) V_r^* v \in \text{Dom } \mathcal{K}_r.$$

This establishes the domain assertion in (4.2) before an inverse is used.

For a positively oriented curve, the outward normal is $-i\gamma_r'/|\gamma_r'|$. A direct calculation therefore gives

$$|\gamma_r'| \partial_\nu u_a = i\partial_\theta h - k\gamma_r' f_0, \quad \gamma_r' f_0 = \mathcal{O}_r v - \frac{ik}{2} \mathcal{T}_r h.$$

Nonresonance identifies u_a with the solution defining Λ_r . Using (4.9), we obtain

$$\Lambda_r h = i\partial_\theta h + \frac{ik^2}{2} \mathcal{O}_r (I - V_r)^{-1} V_r \mathcal{O}_r^* h + \frac{ik^2}{2} \mathcal{T}_r h.$$

All inverses act on the ranges just established. On this domain,

$$i(I - V_r)^{-1} V_r = \frac{1}{2} \mathcal{K}_r - \frac{i}{2} I.$$

Equation (4.5) proves (4.7).

For the orthonormal Fourier basis $(2\pi)^{-1/2}e^{in\theta}$,

$$|\langle h, i\partial_\theta h \rangle| \leq \sum_{n \in \mathbb{Z}} |n| |h_n|^2 \leq \|h\|_{H^{1/2}(\mathbb{T})}^2.$$

The operator $i(\mathcal{T}_r - \mathcal{T}_r^*)$ is bounded and self-adjoint on L^2 , with locally uniform bounds given by its kernel. This proves (4.8), including at resonant radii, since no Dirichlet resolvent occurs in $\mathfrak{r}_r$. $\square$

The last term in (4.7) is asserted only on Herglotz traces. Density alone would not justify extending a composition with the unbounded operator $\mathcal{K}_r$ to all of $H^{1/2}(\mathbb{T})$.

5. DIRICHLET EIGENVALUES AND UNITARY EIGENPHASES

We first record the sign and multiplicity of the singularity of Λ_r as a Dirichlet eigenvalue is approached from smaller domains. The proof uses finite-dimensional spectral projections rather than a choice of varying eigenfunctions.

Lemma 5.1. *Suppose k^2 is a Dirichlet eigenvalue of Ω_{r_*} of multiplicity m , where $r_* > 0$. There are linearly independent $h_1, \dots, h_m \in C^\infty(\mathbb{T})$ and numbers $M_r \rightarrow +\infty$ as $r \uparrow r_*$ such that, for every sufficiently close nonresonant $r < r_*$,*

$$(5.1) \quad \langle h_c, \Lambda_r h_c \rangle \leq -M_r |c|^2, \quad h_c = \sum_{j=1}^m c_j h_j, \quad c \in \mathbb{C}^m.$$

Proof. Every ordered eigenvalue $\lambda_j(\Omega_r)$ is strictly decreasing in r . Indeed, suppose $r < s$ and the two j th eigenvalues equal λ . The first j eigenfunctions on Ω_r , extended by zero, span a trial space on Ω_s with Rayleigh quotients at most λ . It contains a nonzero vector orthogonal to the spectral subspace of Ω_s below λ , whose dimension is at most $j - 1$. The spectral theorem forces that vector to be an eigenfunction on Ω_s at λ . It vanishes on the nonempty open set $\Omega_s \setminus \overline{\Omega_r}$, contrary to unique continuation.

These eigenvalues are also continuous for $r > 0$: use the min-max principle on the fixed disk with denominator $\int_{\mathbb{D}} w_r |u|^2$. On a compact positive-radius interval the weights converge uniformly and are bounded away from zero. By conformal invariance of the Dirichlet energy, the number of negative eigenvalues of A_r in (4.6) is $\#\{j : \lambda_j(\Omega_r) < k^2\}$, and its nullity is the multiplicity of k^2 .

The bounded potentials in A_r vary continuously in L^∞ , so the operators vary in norm resolvent. Let P_r be the Riesz projection of the spectral cluster near zero that equals $\ker A_{r_*}$ at r_* . Its rank is m , and $P_r \rightarrow P_*$ in operator norm. Strict monotonicity and continuity of the domain eigenvalues show that the negative index below r_* equals the negative index at r_* . The spectrum outside the cluster is uniformly separated from zero. Thus all m cluster eigenvalues are positive for $r < r_*$ close enough, and

$$0 < A_r|_{\text{ran } P_r} \leq \varepsilon_r I, \quad \varepsilon_r \rightarrow 0.$$

The inverse on $\text{ran}(I - P_r)$ is uniformly bounded. This argument does not require w_r to be pointwise monotone.

Choose an orthonormal basis $\phi_1, \dots, \phi_m$ of $\ker A_{r_*}$ and put $g_j = \partial_\nu \phi_j|_{\mathbb{T}}$. Elliptic regularity gives $g_j \in C^\infty(\mathbb{T})$. They are linearly independent by uniqueness for zero Dirichlet and Neumann data. In this setting that uniqueness follows by transferring the equation to Ω_{r_*} and extending the solution by zero: both vanishing traces make the extension solve the Helmholtz equation distributionally on the plane, so its Fourier transform, an L^2 function supported on $\{|\xi| = k\}$, must vanish.

Set $h_j = g_j$ and choose fixed smooth extensions $\mathcal{E}h_j$ to $\overline{\mathbb{D}}$. Extend $\mathcal{E}$ linearly on their span. For $h = h_c$ set $f_{r,h} = \mathcal{A}_r \mathcal{E}h \in L^2(\mathbb{D})$, $v_{r,h} = A_r^{-1} f_{r,h}$, and $u_{r,h} = \mathcal{E}h - v_{r,h}$. Then $u_{r,h}$ solves $\mathcal{A}_r u_{r,h} = 0$ with boundary value h . If

$$q_r[p, q] = \int_{\mathbb{D}} (\nabla \bar{p} \cdot \nabla q - k^2 w_r \bar{p} q),$$

Green's identity gives $q_r[v_{r,h}, u_{r,h}] = 0$ and $\langle h, \Lambda_r h \rangle = q_r[u_{r,h}, u_{r,h}]$. Since $\mathcal{E}h = u_{r,h} + v_{r,h}$ and $q_r[v_{r,h}, v_{r,h}] = \langle f_{r,h}, A_r^{-1} f_{r,h} \rangle$, we obtain

$$(5.2) \quad \langle h, \Lambda_r h \rangle = \int_{\mathbb{D}} (|\nabla \mathcal{E}h|^2 - k^2 w_r |\mathcal{E}h|^2) - \langle f_{r,h}, A_r^{-1} f_{r,h} \rangle.$$

At r_* , a second application of Green's identity yields

$$\langle \phi_i, f_{r_*,h} \rangle = \int_{\mathbb{T}} \bar{g}_i h \, d\theta.$$

For $h = \sum c_j g_j$, these coefficients are the product of the positive-definite Gram matrix of the g_j with c . Continuity gives $\|P_r f_{r,h_c}\| \geq c_0 |c|$ for a fixed $c_0 > 0$ and nearby r . The cluster contribution to the last term of (5.2) is at most $-c_0^2 |c|^2 / \varepsilon_r$. The remaining terms are bounded in absolute value by $C|c|^2$. Taking $M_r = c_0^2 / \varepsilon_r - C$ proves (5.1). $\square$

We now use the boundary identity to transfer these negative directions to the spectrum of the Cayley transform.

Proposition 5.2. *Under the assumptions of Lemma 5.1, for every $\pi < \alpha < 2\pi$ and every sufficiently close nonresonant $r < r_*$, the operator U_r has at least m eigenvalues with arguments in $(\alpha, 2\pi)$, counted with multiplicity.*

Proof. Let $J_0 : \mathbb{C}^m \rightarrow H^{1/2}(\mathbb{T})$ be the map $J_0 c = h_c$. At each fixed nonresonant r , Lemma 4.1 provides Herglotz traces $\tilde{h}_{r,j}$ such that the map $J_r c = \sum c_j \tilde{h}_{r,j}$ satisfies

$$\|J_r - J_0\| \leq \delta_r \leq 1, \quad \|\Lambda_r\| \delta_r (2\|J_0\| + 1) \leq 1.$$

Here $\|\Lambda_r\|$ is the norm from $H^{1/2}$ to $H^{-1/2}$. The quadratic-form error is at most $|c|^2$, so (5.1) gives

$$\langle J_r c, \Lambda_r J_r c \rangle \leq -(M_r - 1)|c|^2, \quad \|J_r c\|_{H^{1/2}} \leq C_0 |c|,$$

with C_0 independent of r . The approximation is chosen separately at each radius; no uniform bound on the Herglotz densities is needed.

Set $x_{r,c} = \mathcal{O}_r^* J_r c$. Proposition 4.3 places these vectors in $\text{Dom } \mathcal{K}_r$. Equations (4.7), (4.8), and (4.4) give

$$\langle x_{r,c}, \mathcal{K}_r x_{r,c} \rangle \leq -\frac{4}{k^2} (M_r - C_1) |c|^2, \quad \|x_{r,c}\| \leq C_2 |c|,$$

with constants uniform near r_* . When $M_r > C_1$, this map is injective, since a zero image would contradict the strict negative bound. Its image is an m -dimensional subspace of $\text{Dom } \mathcal{K}_r$ on which the Rayleigh quotient is at most $-4(M_r - C_1)/(k^2 C_2^2)$.

For any $D > 0$, the spectral projection of $\mathcal{K}_r$ onto $(-\infty, -D)$ therefore has rank at least m when r is close enough to r_* . Otherwise that image subspace would contain a nonzero vector orthogonal to the projection, with quadratic form at least $-D$ times its squared norm, a contradiction. Choose D so that $-D < \cot(\alpha/2)$, and use (4.3). $\square$

Only Herglotz waves have been differentiated to form infinite sequences. No such sequence has been assigned to an arbitrary Dirichlet eigenfunction, and no eigenvalue exactly equal to 1 is asserted at a resonant radius.

6. COUNTING AND COMPLETION OF THE PROOF

6.1. Domains with analytic boundary. We retain the spectral tail in the counting argument. The energy parameter in $L_{r,k}$ is written explicitly from this point onward.

Proposition 6.1. *Let $\Omega = F(\mathbb{D})$, where F is holomorphic and injective on a neighborhood of $\overline{\mathbb{D}}$. For every $k > 0$ and every integer m with $1 \leq m \leq N_\Omega(k^2)$,*

$$(6.1) \quad 2\pi m \leq \int_0^1 \Phi_m(L_{r,k}) dr,$$

and

$$(6.2) \quad k^2 |\Omega| \geq 4\pi m + 2 \int_0^1 \tau_m(L_{r,k}) dr > 4\pi m.$$

Proof. Continuity and strict monotonicity of $\lambda_j(\Omega_r)$ were proved in Lemma 5.1. Since $w_r \leq Cr^2$ near zero, the disk Poincaré inequality gives $\lambda_1(\Omega_r) \rightarrow \infty$ as $r \downarrow 0$. It follows that there are finitely many radii $0 < r_1 < \dots < r_q \leq 1$ where k^2 is a Dirichlet eigenvalue. If their multiplicities are $m_1, \dots, m_q$, then

$$(6.3) \quad \sum_{\ell=1}^q m_\ell = N_\Omega(k^2).$$

Indeed, each ordered eigenvalue at most k^2 at $r = 1$ crosses that value exactly once, and every other eigenvalue stays above it. Under the assumptions of the proposition, $q \geq 1$.

Choose nonresonant radii $a_1 \in (0, r_1)$ and $a_\ell \in (r_{\ell-1}, r_\ell)$ for $\ell \geq 2$. Lemma 2.1(ii) gives a common $\alpha_0 \in (\pi, 2\pi)$ such that no $U_{a_\ell, k}$ has spectrum with argument in $[\alpha_0, 2\pi)$. Fix $\alpha \in (\alpha_0, 2\pi)$. Proposition 5.2 provides $b_\ell \in (a_\ell, r_\ell)$ at which there are at least m_ℓ eigenvalues with arguments in $(\alpha, 2\pi)$. The selected eigenvalues persist in that open arc under small perturbations of b_ℓ . Since crossings of $e^{i\alpha}$ are isolated, we may also arrange that $e^{i\alpha} \notin \sigma(U_{b_\ell, k})$.

Lemma 2.1(iii) gives at least m_ℓ crossings in each (a_ℓ, b_ℓ) . These intervals are disjoint. By (6.3), the full path has at least $N_\Omega(k^2)$ crossings, and hence at least m . Applying the strengthened estimate (2.4) to this prescribed number gives

$$\alpha m \leq \int_0^1 \Phi_m(L_{r,k}) dr.$$

The right-hand side is independent of α . Letting $\alpha \uparrow 2\pi$ proves (6.1). Equations (2.1) and (3.3) give (6.2); its last inequality follows from (3.4) and continuity. If $r_q = 1$, the argument still uses $b_q < 1$, so the endpoint multiplicity is included. $\square$

6.2. Arbitrary boundary. A limit of strict inequalities need not be strict. We therefore pass the positive spectral tail, rather than only the strict inequality, through the conformal exhaustion. This completes the proof of Theorem 1.1.

Proof. Let $F : \mathbb{D} \rightarrow \Omega$ be a Riemann map and set $\Omega_s = F(s\mathbb{D})$ for $0 < s < 1$. The map $z \mapsto F(sz)$ is holomorphic and injective on a neighborhood of $\mathbb{D}$. The boundary transport for F is defined on each compact radial interval in $[0, 1)$; the constructions on these intervals agree by uniqueness of the differential equations. We denote its generator by $L_{r,k}$ for $0 \leq r < 1$ and $k > 0$.

Fix $j \geq 1$. We first recall spectral convergence under this exhaustion. Domain monotonicity shows that $\lambda_j(\Omega_s)$ decreases to a limit not smaller than $\lambda_j(\Omega)$. Conversely, the min-max principle and the form-core property of $C_c^\infty(\Omega)$ give, for every $\varepsilon > 0$, a j -dimensional subspace $V \subset C_c^\infty(\Omega)$ such that

$$\max_{0 \neq u \in V} \frac{\int_\Omega |\nabla u|^2}{\int_\Omega |u|^2} \leq \lambda_j(\Omega) + \varepsilon.$$

The union of the supports of a basis of V is compact in Ω , so it is contained in Ω_s when s is sufficiently close to 1. The min-max principle on Ω_s then gives $\lambda_j(\Omega_s) \leq \lambda_j(\Omega) + \varepsilon$. Thus, with

$$(6.4) \quad k_s = \sqrt{\lambda_j(\Omega_s)}, \quad k = \sqrt{\lambda_j(\Omega)},$$

we have $k_s \rightarrow k > 0$ and $|\Omega_s| \uparrow |\Omega|$.

Apply Proposition 6.1 to $F(s \cdot)$ at energy k_s^2 and with $m = j$. This is permitted since $N_{\Omega_s}(k_s^2) \geq j$, including any multiplicity at that energy. Under the radial change of variable $r = st$, the unitary path for $F(s \cdot)$ is U_{st,k_s} and its generator is sL_{st,k_s} . This follows also for the initial-point correction from the defining differential equation for R_r . Since τ_j is positively homogeneous, the proposition gives

$$(6.5) \quad |\Omega_s| \lambda_j(\Omega_s) \geq 4\pi j + 2 \int_0^s \tau_j(L_{r,k_s}) dr.$$

For every fixed $r \in (0, 1)$, the joint trace-norm continuity in Proposition 3.1 and (2.3) imply

$$\tau_j(L_{r,k_s}) \longrightarrow \tau_j(L_{r,k}) \quad (s \uparrow 1).$$

Choose any sequence $s_n \uparrow 1$. Apply Fatou's lemma on $(0, 1)$ to the nonnegative functions $\mathbf{1}_{(0,s_n)}(r) \tau_j(L_{r,k_{s_n}})$, and use (6.5) and (6.4). We obtain

$$(6.6) \quad |\Omega| \lambda_j(\Omega) \geq 4\pi j + 2 \int_0^1 \tau_j(L_{r,k}) dr.$$

The integral is finite because

$$0 \leq \tau_j(L_{r,k}) \leq \text{tr } L_{r,k}, \quad \int_0^1 \text{tr } L_{r,k} dr = \frac{k^2}{2} |\Omega|;$$

the trace identity follows first on compact radial intervals and then by monotone convergence. The integral in (6.6) is strictly positive. Indeed, $L_{r,k}$ is injective on $\ell^2(\mathbb{N}_0)$ for every $r > 0$, and $\tau_j(L_{r,k})$ is positive and continuous on every compact subinterval of $(0, 1)$. Thus (6.6) proves (1.1).

Finally, let $E > 0$. If $N_\Omega(E) = 0$, then (1.2) is immediate. Otherwise put $M = N_\Omega(E)$. Since $\lambda_M(\Omega) \leq E$, the strict eigenvalue inequality gives

$$4\pi M < |\Omega| \lambda_M(\Omega) \leq |\Omega| E.$$

This proves (1.2) and completes the proof. $\square$

The remainder in (6.6) depends on the domain, the chosen conformal parametrization, and the index. Its positivity is sufficient for the strict inequality; no uniform lower bound in terms of the index alone is asserted here.

7. LEAN FORMALIZATION

Both assertions of Theorem 1.1 have been formalized in Lean in the modular source tree rooted at `RequestProject/Main.lean`. The aggregate file imports the three checked modules `MainBase.lean`, `MainMiddle.lean`, and `MainSpectral.lean`. The declarations `main` and `main_counting` prove (1.1) and (1.2), respectively, for every bounded nonempty simply connected planar domain, without a boundary regularity assumption. The formalization identifies $\mathbb{C}$ with $\mathbb{R}^2$ using the usual Euclidean norm and two-dimensional Lebesgue measure.

The development constructs the real and complex spaces $H_0^1(\Omega)$ from the smooth compactly supported core, and the self-adjoint Dirichlet operator associated with the closed gradient-energy form $q_\Omega[u] = \int_\Omega |\nabla u|^2$. Its positive eigenvalues are defined independently of the smooth-core min-max, using orthonormal eigenfamilies of the compact inverse. Lean proves that the two constructions agree, together with the nondecreasing ordering, divergence to infinity, and correct real and complex multiplicities. Thus the formalized counting function is precisely $N_\Omega(E) = \#\{j \geq 1 : \lambda_j(\Omega) \leq E\}$, with the endpoint included and eigenvalues repeated according to multiplicity.

The source tree is pinned to `leanprover/lean4:v4.35.0-rc2` and the matching `mathlib` revision recorded in the repository's `lake-manifest.json`. The Palomar preparation uses this pinned toolchain; the repository records the exact build and audit commands. The recorded verification uses `lake build +RequestProject.Main`. Axiom audits of both final declarations and the operator and spectral identification theorems report only `propext`, `Classical.choice`, and `Quot.sound`,

the standard principles of Lean’s classical mathematics [10]. They do not depend on `sorryAx` or additional axioms; the proof source under `RequestProject/` contains no `sorry`, `admit`, or `native_decide` proof steps.

The original variational development was produced with the assistance of Aristotle; the operator and spectral identification was developed with the assistance of Codex. The source, pinned dependencies, verification commands, build record, and source checksum are provided in the accompanying GitHub repository.¹

REFERENCES

- [1] F. A. Berezin, *Covariant and contravariant symbols of operators*, Math. USSR-Izv. **6** (1972), 1117–1151.
- [2] D. Colton and R. Kress, *On the denseness of Herglotz wave functions and electromagnetic Herglotz pairs in Sobolev spaces*, Math. Methods Appl. Sci. **24** (2001), 1289–1303.
- [3] J.-P. Eckmann and C.-A. Pillet, *Spectral duality for planar billiards*, Comm. Math. Phys. **170** (1995), 283–313.
- [4] K. Fan, *On a theorem of Weyl concerning eigenvalues of linear transformations. I*, Proc. Natl. Acad. Sci. USA **35** (1949), 652–655.
- [5] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, *Pólya’s conjecture for Euclidean balls*, Invent. Math. **234** (2023), 129–169.
- [6] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, *Pólya’s conjecture for Dirichlet eigenvalues of annuli*, J. Lond. Math. Soc. (2) **113** (2026), e70425.
- [7] P. Freitas and I. Salavessa, *Families of non-tiling domains satisfying Pólya’s conjecture*, J. Math. Phys. **64** (2023), 121503.
- [8] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., reprint, Classics in Mathematics, Springer-Verlag, Berlin, 1995.
- [9] P. Li and S.-T. Yau, *On the Schrödinger equation and the eigenvalue problem*, Comm. Math. Phys. **88** (1983), 309–318.
- [10] The Lean developers, *The Lean Language Reference*, Section 8, Axioms, <https://lean-lang.org/doc/reference/latest/Axioms/> (accessed 3 October 2026).
- [11] G. Pólya, *Mathematics and Plausible Reasoning*, Vols. I–II, Princeton University Press, Princeton, NJ, 1954.
- [12] G. Pólya, *On the eigenvalues of vibrating membranes*, Proc. London Math. Soc. (3) **11** (1961), 419–433.
- [13] H. Weyl, *Das asymptotische Verteilungsgesetz der Eigenwerte linearer partieller Differentialgleichungen (mit einer Anwendung auf die Theorie der Hohlraumstrahlung)*, Math. Ann. **71** (1912), 441–479.

¹<https://github.com/QuanyuTang/polya-dirichlet-simply-connected-formalization-Palomar-version>.
The repository is currently private; access must be granted by its owner.